DESIGNING INTERACTIVE PROCESSES IN RECENT ENVIRONMENTAL SOUND ART: *BLUE GOLD* AND *MEDUSA DREAMING*

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ABSTRACT

Interaction. Why and how do we want to relate to the environment? With whom and for whom? With what tools and media? What impact does our work have on the environment, its soundscape?

In two large scale environmental works, *Blue Gold* and *Medusa Dreaming*, processes of interaction will be identified, interrogating how acoustic space, sound, sculpture, film and creative musical interpretation of composed sound structures intersect. In analysing what is fixed and what is unfixed in the design process we can study different modalities of interaction and interdisciplinary collaboration at all stages of the creative process from the idea to audition. Human/environment/ human/machine, human/human interactive processes are endemic to sound art. These relationships will be identified in *Blue Gold* and *Medusa Dreaming* as they are embodied in aspects of the creative works: the site plan, the temporal and spatial design, sound recording, sound design, images, video, installation components, musical composition, to live performance and dispersion.

1. INTRODUCTION

Blue Gold and *Medusa Dreaming* are two recent works of environmental sound art, which provide excellent case studies for looking at the complex simultaneous interactive processes which may occur in interdisciplinary sound art installation and performance. Both pieces are created to focus on the most pressing environmental issue: water care. Both works interrogate our role in how we shape water care and offer time for reflection as to how we may become more attentive to our involvement. *Blue Gold* (2006-2009-12) is an Australian work commissioned by Charisma, a cello and clarinet duo. Original electro-acoustically modified soundscapes of natural and urban Australian environments are overlayed with composed western tunes representing the immigrant nation and human behaviour. Video correlates, actions with sculptures and improvisatory sound structures are required from the performers who must respond to the actual and fictional elements at the same time. *Medusa Dreaming* (2010…) Turkey, is an international cross-cultural, site-specific work celebrating the watercare of the ancients through surround sound, installation and live performance in the Yerebatan Sarayi, a palatial subterranean 6th Century water cistern in Istanbul. The new cross cultural South-North ensemble was created for this event, Erdem Helvacioglu electro-acoustic guitar and live processing, Natalia Mann Samoan harpist, and Izzet Kizzil, Turkish Percussionist and myself on Tarhu,a bowed spike fiddle, amplified slide whistle, and moving flutes. Although there is a common water theme in both works, each has very different design involving complex interactive relationships and processes in order for the work to be realised.

2. INTERACTIVITY: INTERDEPENDENT RELATIONSHIPS AND PROCESSES

Interactive relationships can be defined as those with a co-dependency on each other. "**Inter**, a prefix means 'between' 'among' 'mutually' 'reciprocally' 'together' ...**interactive**, of or pertaining to things or personas which act on each other." The narrower computer term interactivity in systems has come to mean " an immediate response to commands or data," [1] focusing on methodological techniques, often as ends in themselves. Over the years interactivity has had many applications in sound art as I have published elsewhere over the past two decades.[2] Environmental sound art embraces all types of interactive processes and relationships being firmly focused on the situational frame and environmental intention. Electro-acoustic sound is seen as an agent for this. The aesthetic design of environmental sound art requires clarity for each work and how interactivity is situated in terms of deciding

a) what is fixed, composed, not negotiable for change, e.g. a huge acoustic temple?
b) what interdependence and co-operative processes are set in place, (who and what?).

[1] Macquarie Dictionary, Second edition, 1992, p 915.
[2] See bibliography Bandt publications 1-6.

c) what is left open for adaptation and interpretation by the individual/s be it the performer or audience or swimming fish. (unpredictable, non fixed fluidities)

In *Blue Gold* (BG) and *Medusa Dreaming* (MD*)* the designs include a vast array of inter-relational correlates which determine the aesthetic pathway and journey of each work in different ways in its composition, performance and audition: These can be identified as

**environment/ human*, e.g. situational recording, dreaming (Both works)
**acoustic space/technology live sound*, e.g. hydrophone and sound mediated by the acoustic space (cistern, MD, water tank soundscape BG).
**technologically mediated acoustic space* speaker playback, amplified, filtered equalised sound during performance (BG, MD).
**sound/technology:Studio productions: Sound recordings/ mediated by technology, electroacoustic sound*, Prepared composed multi- channel surround sound files brought to the work by the composer, composer/engineer. These elaborate submixes are compositions in themselves. (BG, MD) .
**acoustic space/performer*
How does the harpist respond in the cold dank slippery water corridors at night? How do the drips and long decay time, cold and tuning problems effect performance? Can she hear and relate to the tarhu /flute player several metres and several seconds delay away from other remote corridors? The acoustic position and proximity of the speaker array (MD)?. How does the clarinettist respond playing in a very dark humid night setting on a thin dais on the lake's edge? (BG)
**soundscape/performer* responses,(BG, MD).
**notation/performer(*BG, MD).
**video/performer/audience* (double projections in sand BG, Jellies videos projected on ancient Basilica walls, MD).
**performer/performer* interactions, acoustic and live-electronics, solos, duets,(BG, MD) quartets (MD).
**speaker/audience* spatial proximity to speakers and Decibel comfort, crowds, live performance mix, (MD, BG).
**performer/audience* visibility audible proximity, (MD,BG).

These processes are conceived and overlaid in the design of each work from the outset in conceptual drawings, maps patch drawings, scores and software spectographs.

3. BLUE GOLD

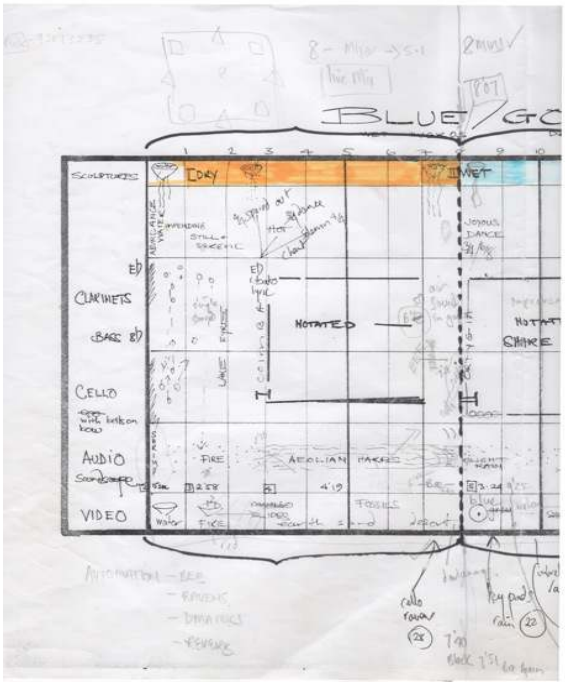


Figure 1. Blue Gold Graphic Score Excerpt.

Blue Gold has a thirty three minute electro-acoustic soundscape to situate the two instrumentalists in wet and dry environments. Their responses to the place of performance influences what and how they play from the composed musical notations and graphic scores. They pour water (cello), and sand (clarinets), into sculptures to tip the balance of the scales, a visual reminder of the state of play in the environment and how we affect the wet/dry relationships we inhabit. A colour -field video is projected behind, or on sand as the case may be, enhancing the wet and arid qualities against the actual environmental setting. In Boreen Point, for the Floating Land Festival curated by Leah Barclay, the video was set as a double projection on the night sand extending to the waters edge. This allowed visual play between the real wet, the lake and the actual dry, the sand. Projecting water on the sand changed this tango. Illumination and back lights further out to sea enhanced this depth of relationships as silver fish jumped from time to time during the performance. The durational aspect of the surround sound soundscape and the video is locked in and is not adjustable from performance to performance, but the elements the performers play can change in the graphic scored sections and in the last wet ,where there is some choice for the quasi improvisatory notation. The performers work with the site and the scored sections to their taste here which makes it very immediate. Sometimes they accompany the soundscape and other times they are submerged by it, or assert themselves as soloists.
The only time in the piece where there is composed rhythmic unison between the performers, the surround soundscape and the sound on the video, is in the cherries sections, which happen twice, representing human greed and destruction of the ecological balance by taking too much. This is fully prescriptive composition with no

room to move by the performers with all the elements fully in synch for the force of capitalism and Western greed to be fully felt. It's the loudest part of the piece. The unison dance *Mazi* (Greek for together), calls for a balance between the wet and dry through working individually and together.

These dynamic relationships may occur and change at any stage of the musical process for the composer, performer, technician, audience. It is the intensity of being in the sound in the environmental location that make the works environmentally present and alive. Not everything can be predicted. Performing *Blue Gold* in arid Wollongong in 2005 inspired Charisma to use the piece as a call for rain to break the drought. In Noosa in 2011 after the Queensland floods, the Boreen Lake performance seemed to celebrate much needed water tranquility. In each locale, the performers are working hard responding to the environments, the tools, maps, musical and graphic notations, listening and responding to the soundscape and each other. They deliver a real-time living work each time *Blue Gold* is played in each venue and particular acoustic space, with its own politic, and the overriding influence of the indoor or outdoor settings. *Blue Gold* exists as a composite of interactive opportunities for each performance. Like the environment, this flexibility of strands and interplays give the work its dynamism. A CD or DVD is merely a postage stamp freezing one of these moments.

4. MEDUSA DREAMING

4.1 Designing and Interacting with an Acoustic Space: the Yerebatan Sarnici, Istanbul. Beginning the designs.

Medusa Dreaming is coming from a very different inspiration. It is a site specific work designed for a particular acoustic space, a world heritage site, the Yerebatan Sarnici, an ancient subterranean water palace, one of the most visited world heritage sites in Turkey. Its architectural composition, acoustic identity its history and function have influenced all aspects of this work. It is 140m long, 70m wide and is a giant rectangular with 336 marble columns reaching 9 metres to support its Romanesque arched roof. 100,000 tonnes of water can be stored in it. Today the carp swim in the waters below the platforms where tourists can walk. There is usually piped music, Handel's water music or the Turkish end blown flute, the Ney coming from speakers. The first of two Saturday evening concerts, *Sounding the Basilica,* the duet with Erdem Helvacioglu, was a sonic investigation of all aspects of this chamber with an array of recordings I prepared, live electronic explosions from Erdem, quiet hydrophone recordings from the fish and improvised interactive music. We investigated how sound behaved in the air and underwater, in close proximity, playing flute through the corridors, catching the nodal spots, up down, natural and electronic, processed, unprocessed. With my hydrophone, live realtime recordings of fish swimming in the water below the platforms, were heard by the audience accompanied by the resonating condensation of drips falling from the high vaulted stone and brick Romanesque ceiling, the tears of Medusa themselves. Medusa jellyfish videos were projected on the ancient walls, the sound track having been prepared at home by playing my glass percussion sculpture, the flagong, improvising from the videos as kinetic scores.

The knowledge of the behaviour of this tricky acoustic was invaluable as sound designer and composer for *Medusa Dreaming*, although I had visited and recorded the ambience on two former occasions. Improvising on the tarhu with the harpist on the Tuesday before the second concert, taught us a lot about our duet for the sixth movement in terms of balance and possible performer interactions.The stage was quite dead and contained by comparison of working freely in the corridors.

4.2 Pre-performance Processes and Interactive Methods: composer, technician, musician interactions, Melbourne.

The surround sound files for each movement were composed and spatialised in the ABC studios beforehand together with engineer Alex Stinson. Key words inspired from the space, "water", "turned to stone", "336 marble columns" were recorded in international languages, Italian, Ancient Greek, French, Thai, Somali, Spanish, Kurdish, and positioned spatially so they would emerge from behind the columns throughout the space, as though the building itself were talking. 52 glass bells were struck representing the 52 steps everyone would climb at the end of the piece when leaving. An enormous palette of original samples was created, including pitch shifted glass bells, granulated bass slide whistle, and original water bottle sprung sculptures, bowed and played.

For the water harps track, psaltery and autoharps were performed and recorded in Melbourne together with my former sound art students, Angie Grant and Albert Mishriki. Angie treated the psaltery and autoharp improvisations with effects and created multi channeled sub-mixes as composed elements for particular movements in surround.

4.3 Composer Working with Turkish Musicians

Medusa Dreaming is a structured electroacoustic symphony in twelve movements for an international north south ensemble I arranged, Natalia Mann Samoan harpist, Erdem Helvcioglu live electroacoustic guitar and effects, Izzet kizzil, Turkish percussionist and Ros Bandt on slide whistle, bowed spike- fiddle, the tarhu, renaissance flute, and Pythagorean recorder.

Each performer was sent a page of composed notation which could be the tuning mode and modules for the improvised pieces.



Figure 2 Medusa Dreaming Notation

4.4 Medusa Dreaming's Structural Design

The programme was designed so that the performers were assigned particular roles in each movement. The lead follow, duet, exchange, support, accompany was broadly outlined as a common plan and discussed and rehearsed.

I *The Tears of Yerebetan Palace*: walking flute prelude, articulating the room, small sounds.

II *Frozen locks*: Athena's curse, (cello guitar/processing) drying airwhistle, ancient Greek "Turned to Stone " words,

III *Fish Dreaming*: fish solo enhanced hydrophone with percussion.

IV *Ode to Emperor Justinianus:* Ottoman duet, no soundscape, live electronics, Turkish Male duet.

V *Water through Glass:* Glass modules, harp water, live electronics, flute and harp.

VI *Corinthian Song* recorder percussion MD, theme introduced and built by the 4tet to rhythm 4.13

VII *Water Dreaming* water words over soundscape, water harps, flute tune and quartet accompaniment.

VIII *Doric Song* mediaeval dance, flute perussion.

IX *Medusa Dreaming* Harp takes the MD Notation with percussion, others accompany. Natalia's Medusa Solo.

X *Basilica Dreaming* electro-acoustic columns of words over aeolian harps soundscape with cello guitar.

XI *From Belgrat Forest,* woods rimu tree growing Tarhu and cello guitar.

XII *52 steps to the future of water*, all international voices, water soundscapes, improvisation by quartet.
Harp, air whistle, cello guitar /effects, percussion, apotheosis to stairs.

A CD of the electro-acoustic sound files I prepared for each movement with individual instructions was sent to all the performers beforehand. Each person was invited to make a personal response to the Medusa Myth. There would be 2 rehearsals and a long radio sound check in the space, the day of the concerts.

I made a situation which would be fun for all the professionals to enjoy the space and interaction in a public concert setting to inspire the audience to reflect on the history of the cistern in Istanbul through the ages, Doric, Corinthian, Ottomon and Modern references having been planned with the surround sound files.

4.5 Engineering Design

The ABC and TRT Radio stations agreed to provide the recording between them. I had already prepared the materials for playback. In Istanbul I made a hydrophone recording of the carp fish swimming in the cistern waters. This was mixed for a solo to entice the audience to listen just to the space with the drips near the beginning of the piece as a way in to "hearing the Cistern itself". Erdem's post-production enhanced the spatialisation of the fish from underwater to the air. This was added to the soundscape movement. Erdem would have his own live electronics set up onstage, and would oversee the soundscape files playback I had prepared as well as his own patches. This ensured integration of the material. The TRT would have a different desk which took the hydrophone line in as well. My Turkish engineering assistant Isik Yilmaz oversaw this interface, and was the translator and runner between. There was lots of essential co-operation by a team of professionals minding the sound at a number of levels in this vast and tricky space.

The multiple set ups, fold back, hydrophone live feeds, 12 microphones, and three mixing desks had to serve the peromrs, the audience and the radio documentation at the same time.

4.6 Contrapuntal processes

Experiencing the full gamut of these relationships and interactive processes in such an acoustic space was very powerful for all of us, the audience, Radio and TV, as the immediacy of having such an international event with so many contributors involved in the making of it, meant the world came into the space and was celebrating water together. The grandeur of this water palace was celestial, dense, dark and radiant, alchemical. Its overpowering strength as an acoustic modifier, a fish habitat, an elevated watery theatre, and psychological filter can not be underestimated, nor should it be. Medusa Dreaming is a celebration of the care the

ancients took with water tanks but also the references to *Dreaming* allude to the Australian aboriginal sustainable water-care, practices thousands of years before the cistern was even thought of in the Mediterranean. Water wisdom is a basic human necessity, whatever the nation or time.

5. CONCLUSIONS: COMPARISONS AND CONTRASTS

Blue Gold and Medusa Dreaming contain many similar processes, human to human, engineer to composer, performer to performer, performer audience. Both works engage in a complex array of human-machine/instrument engagements which are varied in each movement of the two works, live electronics, computerised heavily processed sound samples, derived from both natural, and original sculptural sources, live processing, improvising with surround soundscapes, shared musical performance.
The recording process of the entire work of course exists in the computer,before, during and after. In Australia, the ABC carried out post production on the TRT files I brought back, a truly international collaboration.

The engagement with the site is paramount in both works, even though in Medua Dreaming it is fixed and in BG it changes each performance. The cistern inspires, but so too does the natural environment. However Medusa Dreaming would be a very different work if it were to be performed in another venue such as the outdoor Federation Square in Melbourne. A variety of technological systemic interactions exist in the preparation of the surround sound fields both in the studio and in the live electronic performance patches.
In both works the performers have to decode instructions, read notation, improvise from modules and make decisions about intensity of self /other relationships. All the notated sections in the pieces are inspired by the Mediterranean and all the soundscapes are from Australian recordings except for the live fish recording from the waters of the cistern itself (MD).
In both works it is the contrapuntal fluidity of these changing interactions which gives each work its dynamism. Environmental water awareness is encouraged through sounding the different spaces which provide opportunities for attentive and reflective listening for all.

6. REFERENCES

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